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Supplementary Data for Layered representation diagnostics for short metagenomic reads

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Abstract

Supplementary figures and source tables supporting the representation-diagnostic analyses, including baseline, estimator, perturbation, parameter, redundancy and runtime audits.

Keywords supplementary data, metagenomics, representation diagnostics

Supplementary data overview

The canonical 4-mer, property and multi-scale property-pooling representation (CK4P-MSP) contains a canonical 4-mer composition block (K), a global property block (P) and multi-scale property pooling (MSP). This supplementary file collects the visual audits and source-table previews that support the main representation-diagnostic manuscript. Figures S1–S3 evaluate baseline controls, empirical information proxies and sequencing-error-aware perturbations. Figures S4–S7 examine local mutation fraction, P-channel counterfactuals, MSP bin and weight sensitivity, and dimension-matched high-k compressed baselines. Figures S8–S10 summarize P/MSP contribution and relation audits and the CAMI II marine anonymous-read stability probe. Figure S11 reports the mechanism-aligned 2×2 local-change audit and property-scaling sensitivity. Figure S12 compares CK4P-MSP with representative historical handcrafted descriptors by drift, grouped local-change readability, dimension and common-protocol implementation time. Figure S13 reports the dense 50–75 bp continuity audit, and Figure S14 separates frozen-head representation geometry from downstream scaling and regularization. Tables S1–S9 provide the CAMI II preview, seven-group contrasts, MI inference, factorial contrasts, scaling sensitivity, complete feature definitions, public WGS assembly accessions, cross-target fixed-head results and CAMI target-task metadata. Unless a caption explicitly defines an interval, bars, points and lines show analysis-cell means without error bars. Cell-bootstrap intervals summarize variation across pre-specified matched length-by-condition or length-by-local-mode cells; because these cells reuse source template panels, they are not confidence intervals for independent biological cohorts.

Supplementary Figures

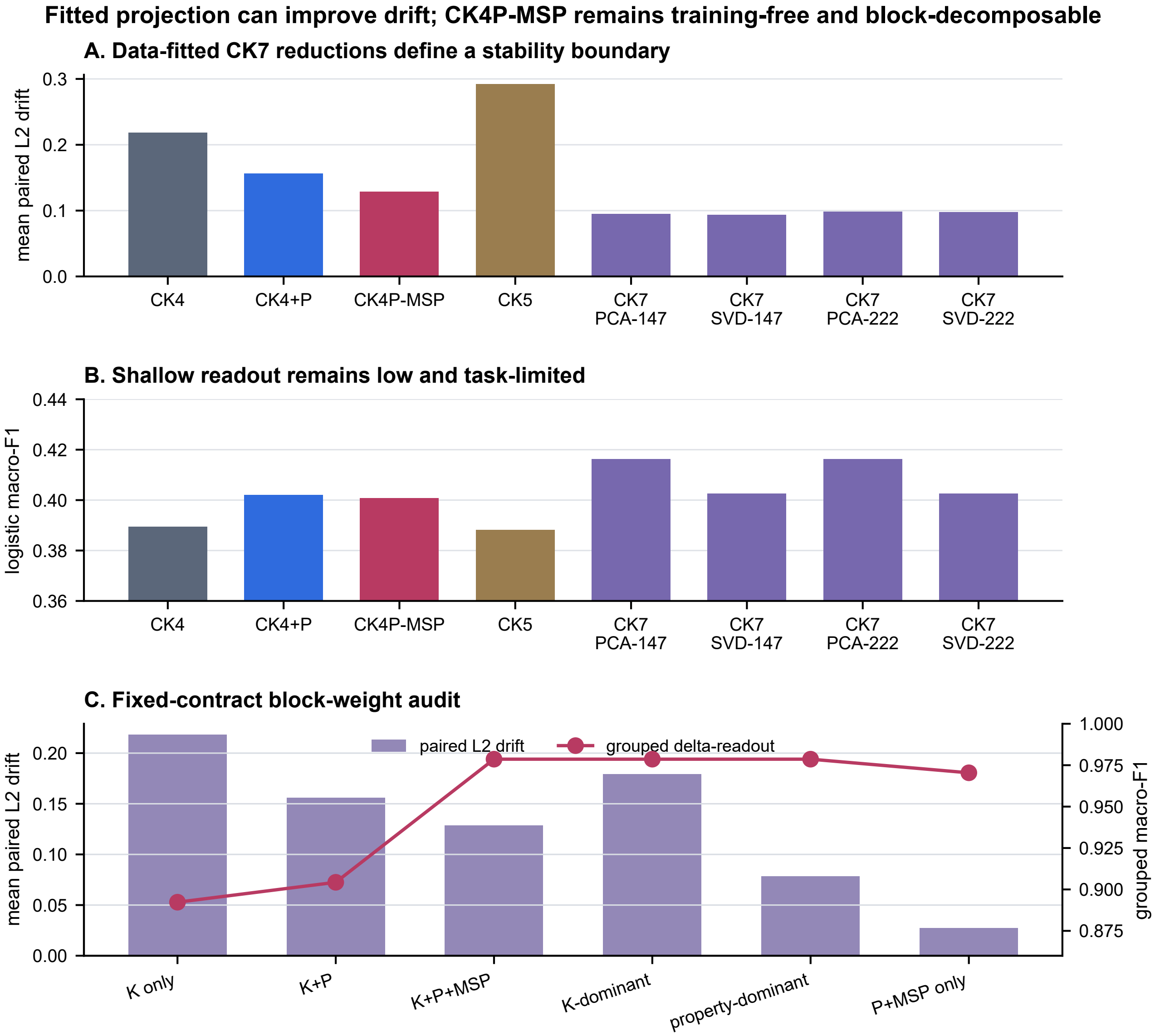

Figure S1 Fitted-reduction and block-weight boundary audit. (A) Mean standardized representation drift across 24 controlled WGS length-by-perturbation cells for CK4/CK5, CK4+P, CK4P-MSP and clean-partition-fitted CK7 PCA/SVD projections at 147 or 222 target dimensions; lower is preferable. The fitted CK7 projections achieve lower drift than CK4P-MSP and therefore define a data-dependent stability boundary. (B) Mean logistic macro-F1 across 132 WGS readout cells; projection rank in these smaller per-cell training partitions was sample-limited (median 49.5 components), so this panel is a shallow-readout boundary rather than a strict 147/222-dimensional comparison. (C) Mean drift across 24 WGS cells and grouped local-change macro-F1 across 12 cells under K-only, K+P, K+P+MSP and prespecified K/property-dominant weights. Bars, points and lines are cell means without uncertainty intervals. The identical grouped readout across the three non-zero K/P/MSP weightings shows that the default is not a fragile readout optimum.

Alt text: Three panels show that fitted CK7 PCA/SVD projections have lower drift than CK4P-MSP, all shallow readouts remain low, and the grouped local-change result is stable across non-zero K/P/MSP weightings.

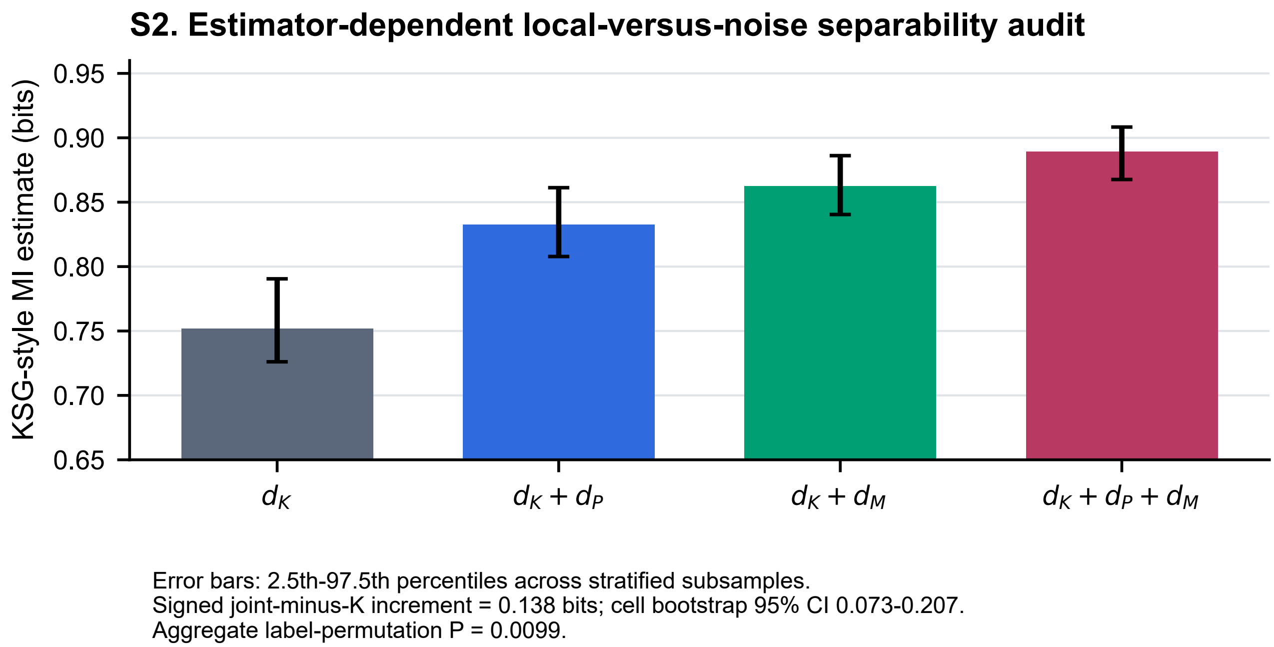

Figure S2 Estimator-dependent mutual-information (MI) audit for local change versus matched nuisance perturbation across 12 length-by-local-mode cells. Bars show KSG-style estimates for d_K, $d_K + d_P$, $d_K + d_M$ and $d_K + d_P + d_M$; error bars are the 2.5th–97.5th percentiles across 100 stratified subsamples per cell. The signed joint-minus-K increment was evaluated with a 10,000-resample paired-cell bootstrap and 100 shared label permutations per cell. Values are empirical separability estimates rather than universal information bounds; higher values indicate greater local-versus-nuisance separability under this estimator.

Alt text: Bar plots summarizing empirical mutual-information and conditional-mutual-information audit values across representation blocks.

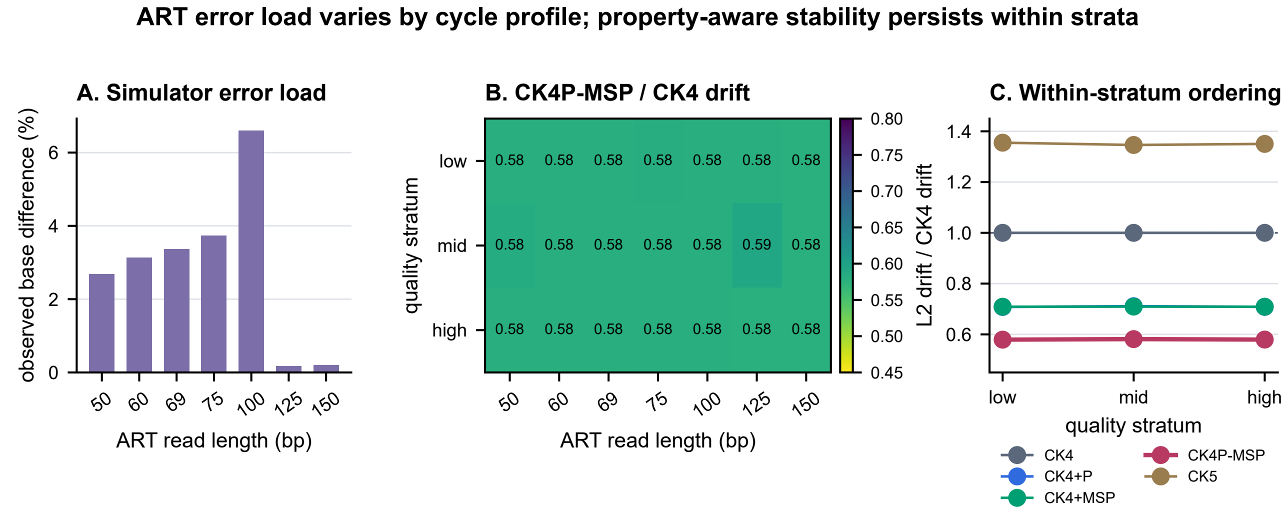

Figure S3 Current-contract ART error-load and quality-stratification audit using 55,961 matched clean/simulated pairs, represented as 111,922 rows, at 50, 60, 69, 75, 100, 125 and 150 bp. (A) Bars show the observed same-coordinate base-difference rate at each length; this descriptive mismatch rate is not an alignment-aware edit distance. (B) Heatmap cells show CK4P-MSP standardized L2 drift divided by CK4 drift within each length and low-, middle- or high-quality stratum; values below one indicate less drift than CK4. (C) Lines show mean within-stratum drift ratios for CK4, CK4+P, CK4+MSP, CK4P-MSP and CK5, averaged over the seven lengths. Panels report descriptive means without uncertainty intervals. Because the selected ART cycle profile produces a non-monotonic error load across lengths, the figure supports within-length and within-stratum ordering rather than a causal read-length trend.

Alt text: Three panels show non-monotonic ART base-difference rates, CK4P-MSP-to-CK4 drift ratios by length and quality, and stable within-stratum representation ordering.

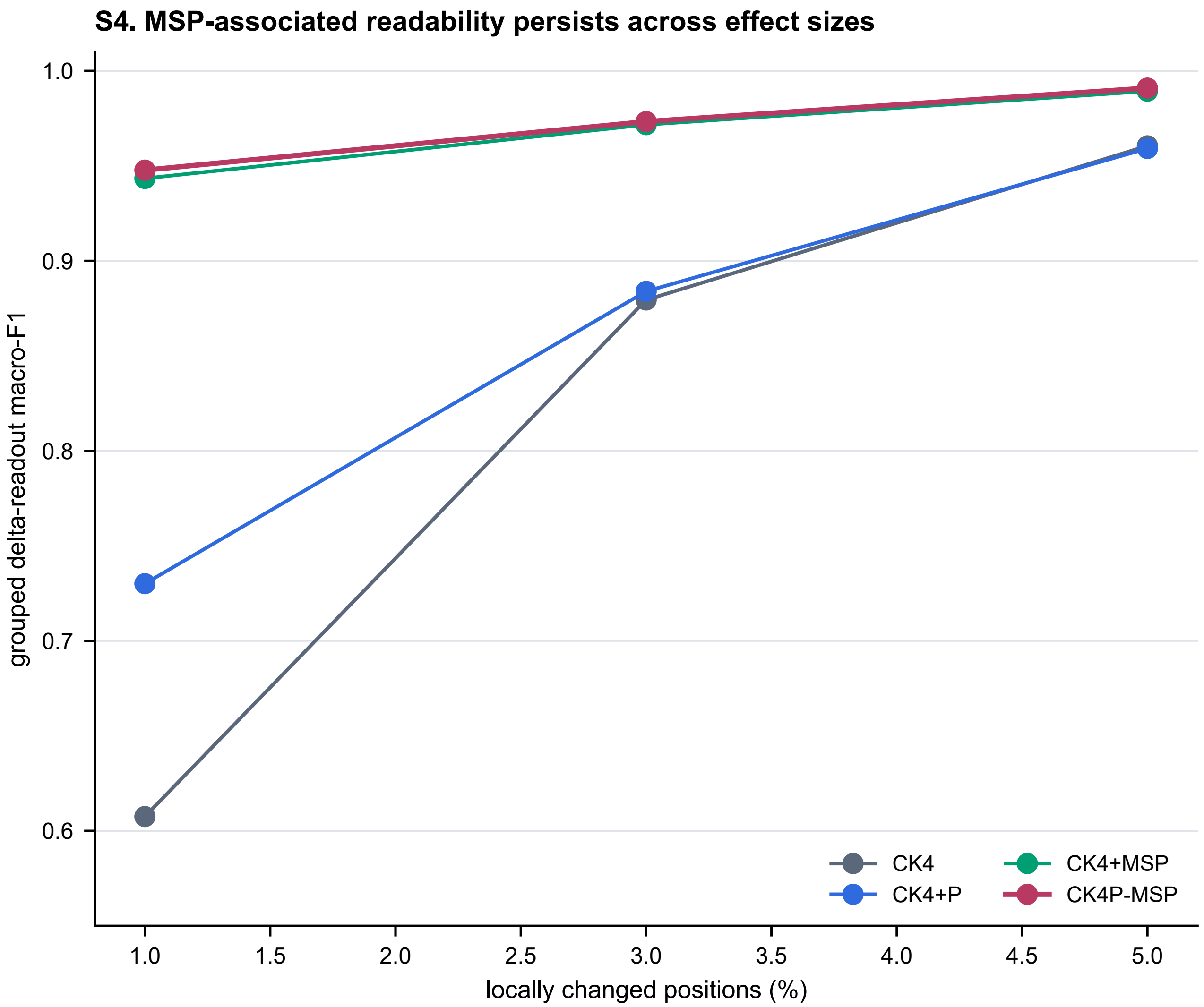

Figure S4 Contract-v2 local-change fraction sweep at 1%, 3% and 5% changed positions. Curves show cell-mean logistic grouped delta-readout macro-F1 without uncertainty intervals, averaged across 12 length-by-local-mode cells with 180 template triplets per cell and all derivatives of one template retained in one fold. Higher values indicate greater accessibility of local-versus-nuisance structure. CK4+MSP and CK4P-MSP retain the strongest readout across effect sizes; this synthetic mechanism probe is not a functional-variant classification task.

Alt text: Line plot showing local mutation-fraction sensitivity across compact representation families.

Supplementary Tables

Table S1. Preview of the current CAMI II marine anonymous-read source table. The 12 rows show the four prespecified compact representations at 69, 75 and 100 bp under 3% N masking; each row summarizes 4,000 clean/perturbed pairs. The complete length-by-perturbation CSV is retained in the data package.

length	condition	method	features	paired cosine	L2 drift	top-1
69	N_3pct	CK4	136	0.9643	0.2649	1.0000
69	N_3pct	CK4+P	147	0.9812	0.1926	1.0000
69	N_3pct	CK4P-MSP	222	0.9871	0.1594	1.0000
69	N_3pct	CK5	512	0.9383	0.3487	1.0000
75	N_3pct	CK4	136	0.9683	0.2497	0.9998
75	N_3pct	CK4+P	147	0.9833	0.1814	0.9998
75	N_3pct	CK4P-MSP	222	0.9886	0.1501	1.0000
75	N_3pct	CK5	512	0.9440	0.3324	1.0000
100	N_3pct	CK4	136	0.9686	0.2486	1.0000
100	N_3pct	CK4+P	147	0.9833	0.1816	1.0000
100	N_3pct	CK4P-MSP	222	0.9886	0.1500	1.0000
100	N_3pct	CK5	512	0.9406	0.3428	1.0000

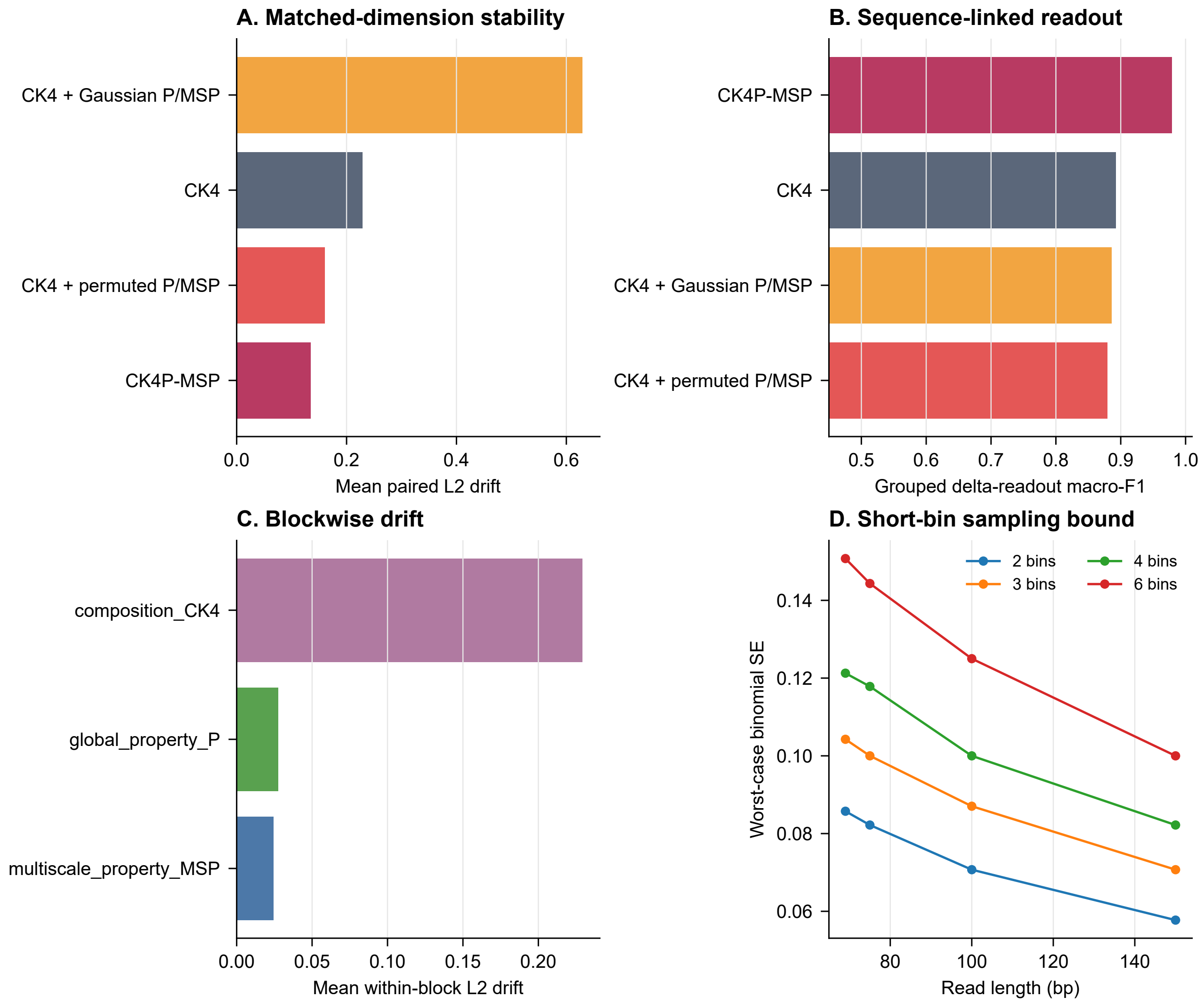

Figure S5 Sequence-linked P/MSP counterfactual and short-bin reliability audit. (A) Mean standardized drift across 20 WGS length-by-perturbation cells compares CK4P-MSP with CK4 and with row-permuted or marginally matched Gaussian P/MSP controls, all under the fixed block-normalization contract. (B) Mean logistic grouped delta-readout across 12 local-change cells (250 template triplets per cell) is higher for sequence-linked CK4P-MSP (0.979) than for CK4 (0.892), permuted P/MSP (0.879) or Gaussian P/MSP (0.886). (C) Cell-mean K, P and MSP blockwise drift separates their perturbation responses. Panels A–C do not show uncertainty intervals. (D) Worst-case binomial standard error is shown for 2-, 3-, 4- and 6-bin scales over the tested WGS anchor lengths, based on the shortest bin at each scale. Lower drift and sampling error and higher readout are preferable on their respective panels.

Alt text: Multi-panel audit of P-channel counterfactual behaviour, drift decomposition and short-bin sampling bounds.

Static MSP sensitivity across representative WGS conditions and the 50 to 75 bp sweep

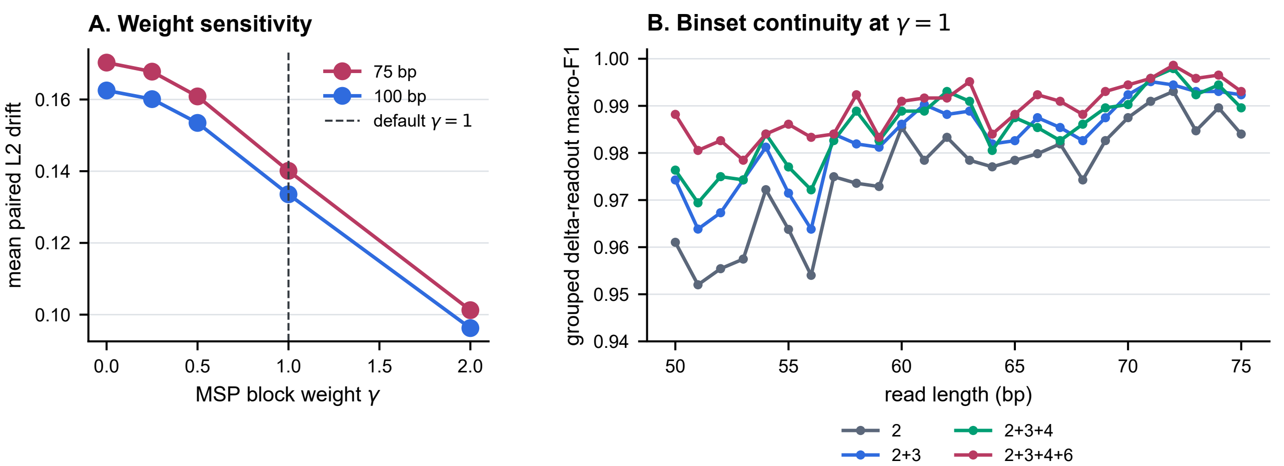

Figure S6 MSP binset and weight-sensitivity audit. (A) Points are mean drift without uncertainty intervals over the prespecified WGS perturbations at 75 and 100 bp for the complete 2+3+4+6 binset and $\gamma \in \{0, 0.25, 0.5, 1, 2\}$; lower is preferable. (B) Lines are mean grouped delta-readout macro-F1 over four local modes at every integer length from 50 to 75 bp, using 180 template triplets per length-by-mode cell at $\gamma = 1$ for cumulative binsets 2, 2+3, 2+3+4 and 2+3+4+6; higher is preferable. All derivatives of one template were retained in the same fold during fivefold grouped cross-validation. No uncertainty intervals are shown. The audit tests sensitivity rather than selecting an optimal binset or weight.

Alt text: Line plots show gamma-weight sensitivity at 75 and 100 bp and MSP binset continuity across every length from 50 to 75 bp.

High-k vector controls and MinHash require separate geometries

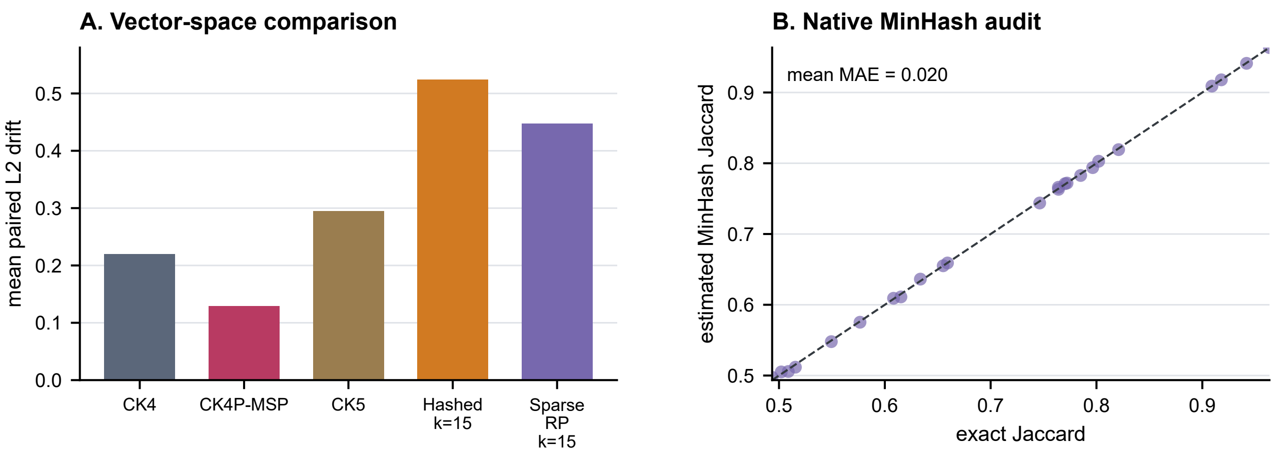

Figure S7 Dimension-matched high-k and native MinHash audit across 24 controlled WGS length-by-perturbation cells. (A) Bars are cell-mean standardized representation drift without uncertainty intervals for hashed and sparse-random-projection canonical $k = 15$ vectors, CK4, CK5 and CK4P-MSP at a 222-feature budget; lower is preferable. (B) Each point compares 222-hash MinHash signature agreement with exact $k = 15$ Jaccard similarity in one matched cell; the diagonal denotes equality and lower absolute error indicates better approximation. MinHash is not treated as an L2 feature vector.

Alt text: Multi-panel audit comparing dimension-matched high-k compressed vectors and native MinHash agreement with exact Jaccard similarity.

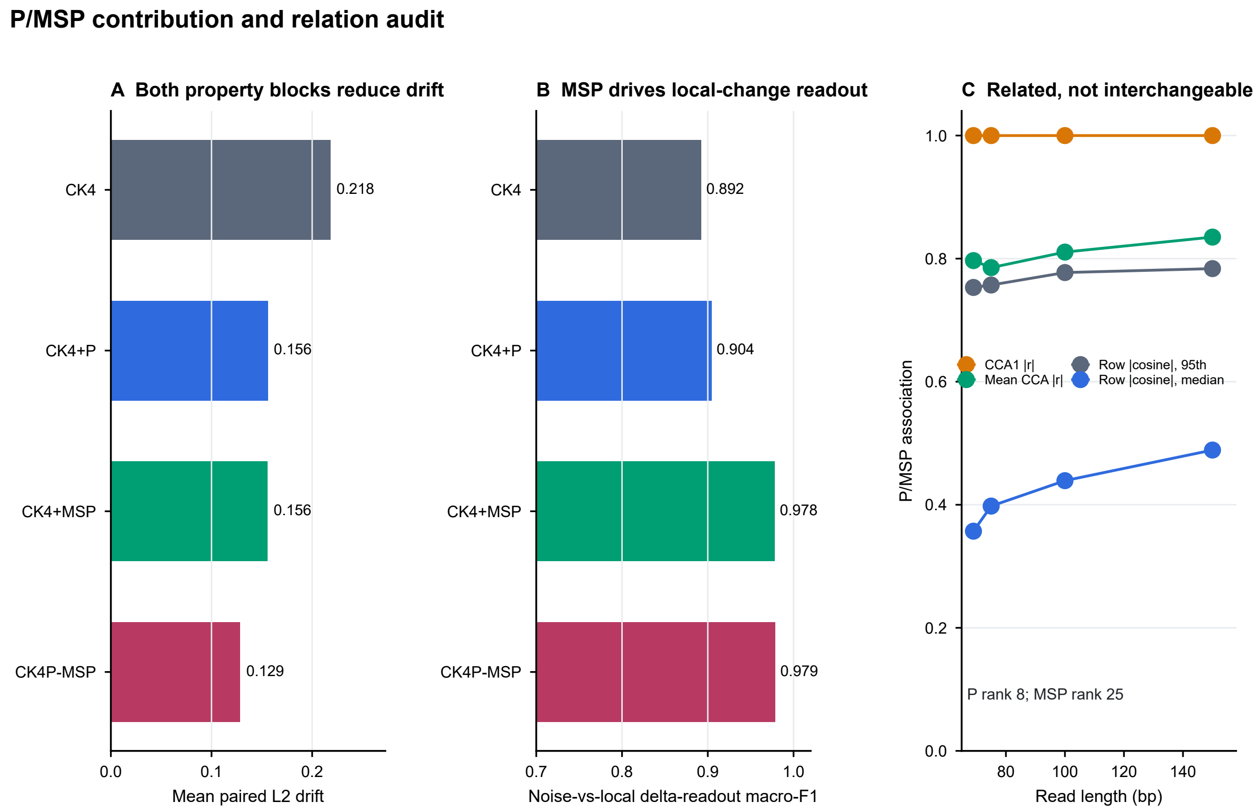

Figure S8 P/MSP contribution and relation audit. (A) Bars are mean paired L2 drift without uncertainty intervals across 24 controlled WGS length-by-perturbation cells for CK4, CK4+P, CK4+MSP and CK4P-MSP; lower is preferable. (B) Bars are mean template-grouped local-versus-nuisance delta-readout macro-F1 without uncertainty intervals across 12 length-by-local-mode cells with 250 template triplets per cell; higher is preferable. (C) Descriptive canonical correlations and row-wise alignment are calculated from 600 clean reads at each of 69, 75, 100 and 150 bp; no inferential intervals are shown. P and MSP share a strong leading biochemical component, whereas incomplete row-wise alignment and different numerical ranks support related-but-non-equivalent wording.

Alt text: Three panels show that P and MSP both reduce paired drift, MSP contributes most of the grouped local-change readout gain, and the two property blocks share leading structure without being row-wise interchangeable.

Table S2. Prespecified conditional-contribution contrasts from the seven-group K/P/MSP ablation. The full representation is CK4P-MSP. Improvement is signed so that positive values favour the full representation; for L2 drift, the sign is reversed because lower values are preferable. Confidence intervals are percentile intervals from 10,000 paired-cell bootstrap resamples. Two-sided Wilcoxon signed-rank tests were adjusted across all 12 rows by the Benjamini–Hochberg procedure. Global cells are length-by-perturbation pairs; local cells are length-by-local-mode grouped cross-validation summaries. Cells reuse source template panels, so the intervals and tests quantify consistency across the prespecified grid rather than independent biological-cohort inference.

block contrast	evidence layer	metric	n	full	comparator	improvement [95% CI]; BH q
K | P+MSP	global	paired cosine	24	0.989	0.999	-0.010 $[-0.013, -0.008]$; 2.4×10^{-7}
K | P+MSP	global	L2 drift	24	0.129	0.027	-0.101 $[-0.118, -0.086]$; 2.4×10^{-7}
K | P+MSP	global	retrieval	24	1.000	0.942	0.058 $[0.036, 0.081]$; 7.3×10^{-4}
K | P+MSP	local	grouped macro-F1	12	0.979	0.970	0.008 $[-0.0002, 0.019]$; 0.507
P | K+MSP	global	paired cosine	24	0.989	0.984	0.005 $[0.004, 0.006]$; 2.4×10^{-7}
P | K+MSP	global	L2 drift	24	0.129	0.156	0.027 $[0.023, 0.031]$; 2.4×10^{-7}
P | K+MSP	global	retrieval	24	1.000	1.000	0.000 $[0.000, 0.000]$; 1.000
P | K+MSP	local	grouped macro-F1	12	0.979	0.978	0.001 $[-0.0002, 0.002]$; 0.525
MSP | K+P	global	paired cosine	24	0.989	0.984	0.005 $[0.004, 0.006]$; 2.4×10^{-7}
MSP | K+P	global	L2 drift	24	0.129	0.156	0.027 $[0.023, 0.032]$; 2.4×10^{-7}
MSP | K+P	global	retrieval	24	1.000	1.000	0.000 $[0.000, 0.000]$; 1.000
MSP | K+P	local	grouped macro-F1	12	0.979	0.904	0.074 $[0.035, 0.117]$; 7.3×10^{-4}

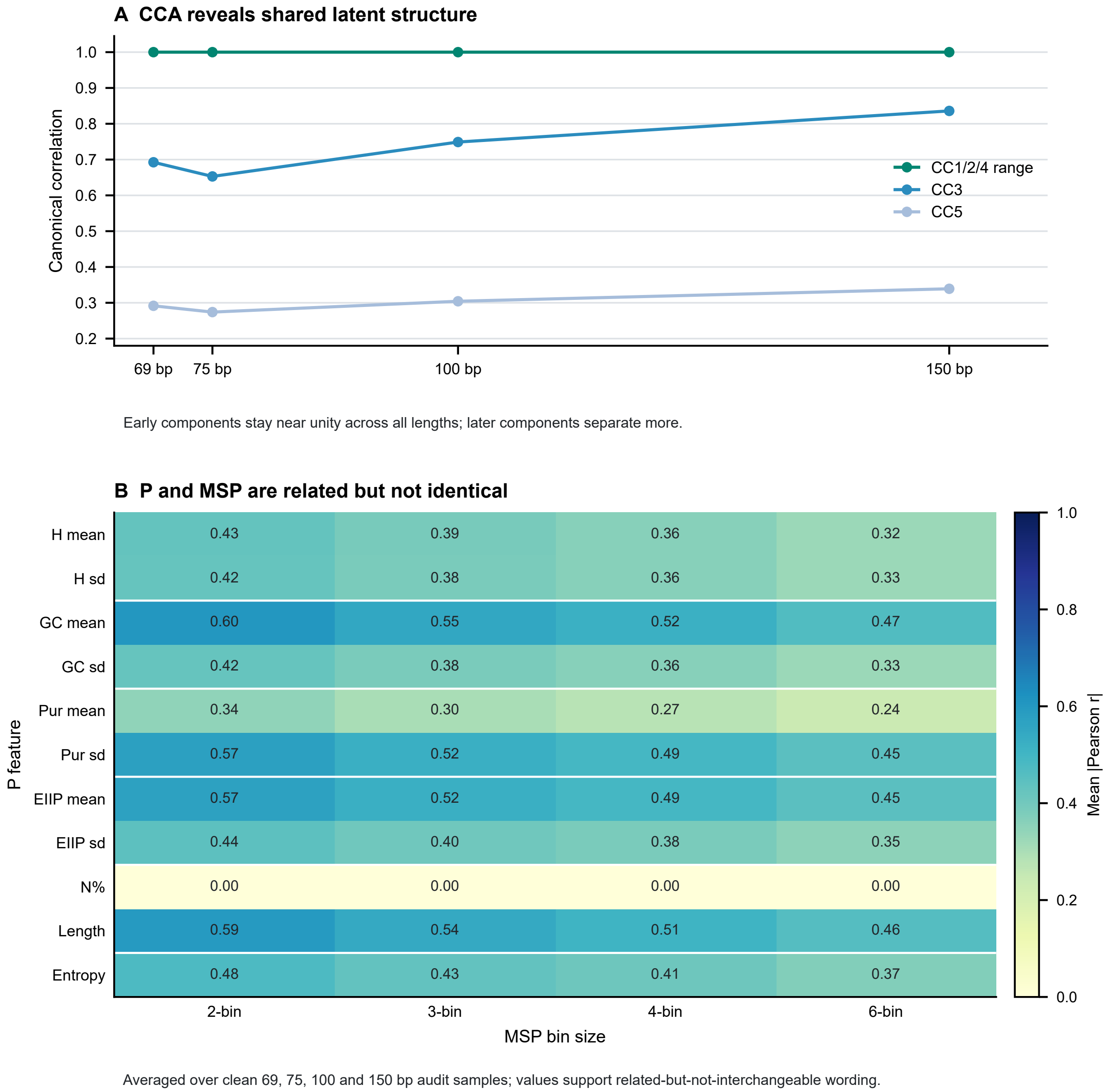

Figure S9 P/MSP relation audit using 600 clean reads at each of 69, 75, 100 and 150 bp. (A) Canonical-correlation analysis summarizes shared latent structure across the 11-dimensional P and 75-dimensional MSP blocks. (B) The heatmap reports mean absolute Pearson correlation between each P coordinate and each MSP bin-scale family, averaged across the four lengths. Values are descriptive estimates without uncertainty intervals. Values near one indicate strong association, not identity; the audit supports related-but-non-equivalent wording and is not a test of statistical independence.

Alt text: Line and heatmap panels showing CCA and correlation relationships between P and MSP feature layers.

Table S3. Estimator-dependent empirical MI separability increment across 12 prespecified length-by-local-mode cells. The contrast is signed and was not clipped at zero. The confidence interval uses 10,000 paired-cell bootstrap resamples; the aggregate permutation P value uses 100 shared label permutations per cell. The interval describes analysis-grid variation and is not an independent biological-cohort confidence interval.

contrast	n cells	mean (bits)	95% CI	Wilcoxon P	permutation P	k
$\widehat{I}(d_K, d_P, d_M; Y) - \widehat{I}(d_K; Y)$	12	0.138	[0.073, 0.207]	0.0004883	0.009901	5

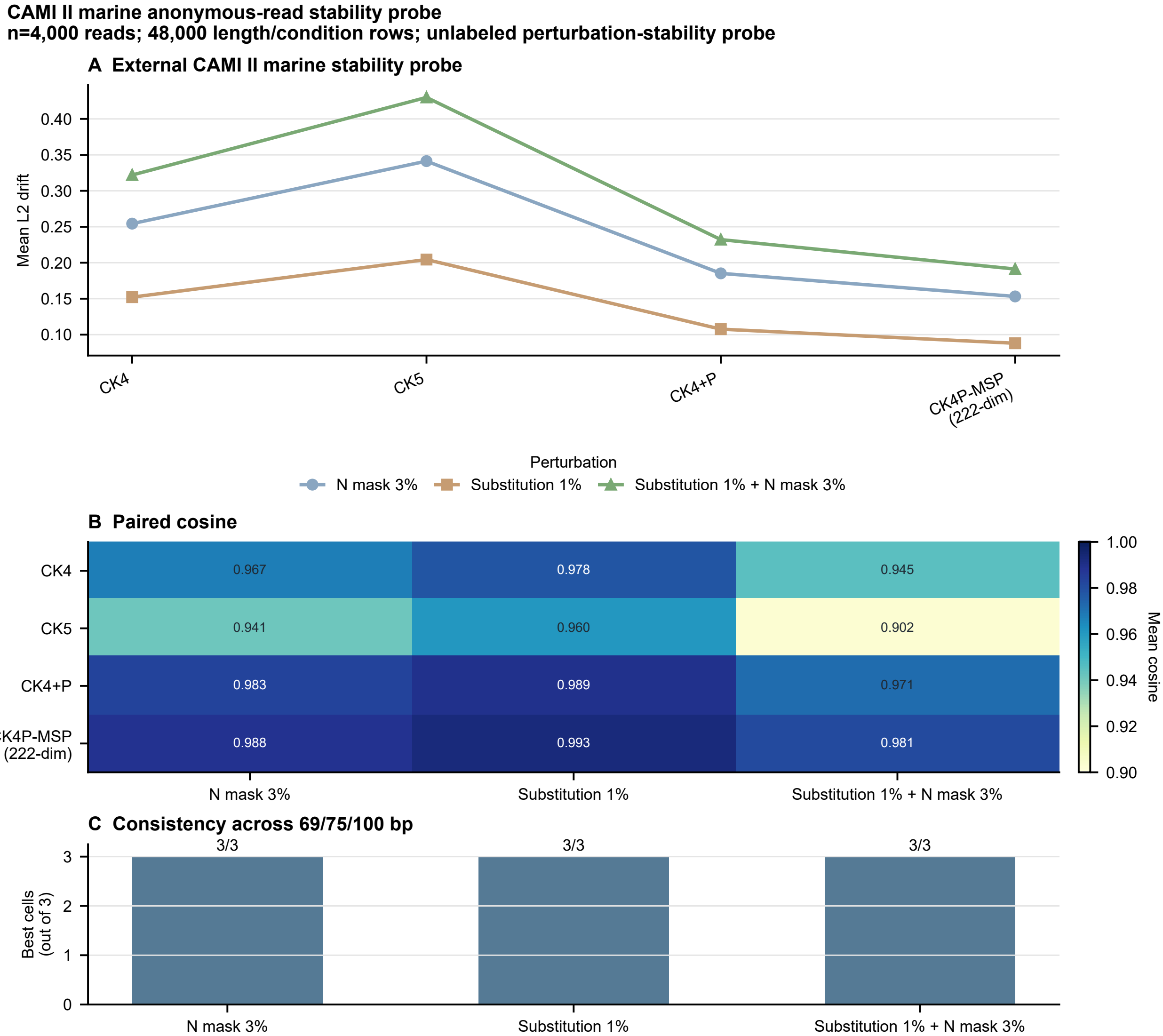

Figure S10 CAMI II marine anonymous-read stability probe using 4,000 anonymous source reads, expanded to 48,000 length-by-condition rows after truncation to 69, 75 and 100 bp and application of N masking, substitution and combined perturbations. (A) Lines show mean standardized L2 drift across the three lengths for each representation and perturbation; lower is preferable. (B) Heatmap cells show mean paired cosine across the three lengths; higher is preferable. (C) Bars count, for each perturbation, the number of length cells out of three in which CK4P-MSP had the lowest L2 drift among the displayed methods. Panels show descriptive cell means or counts without uncertainty intervals. Read-level taxonomic labels were not reconstructed, so this figure supports external stability only.

Alt text: Line, heatmap and bar panels showing CAMI II marine anonymous-read stability across length and perturbation settings.

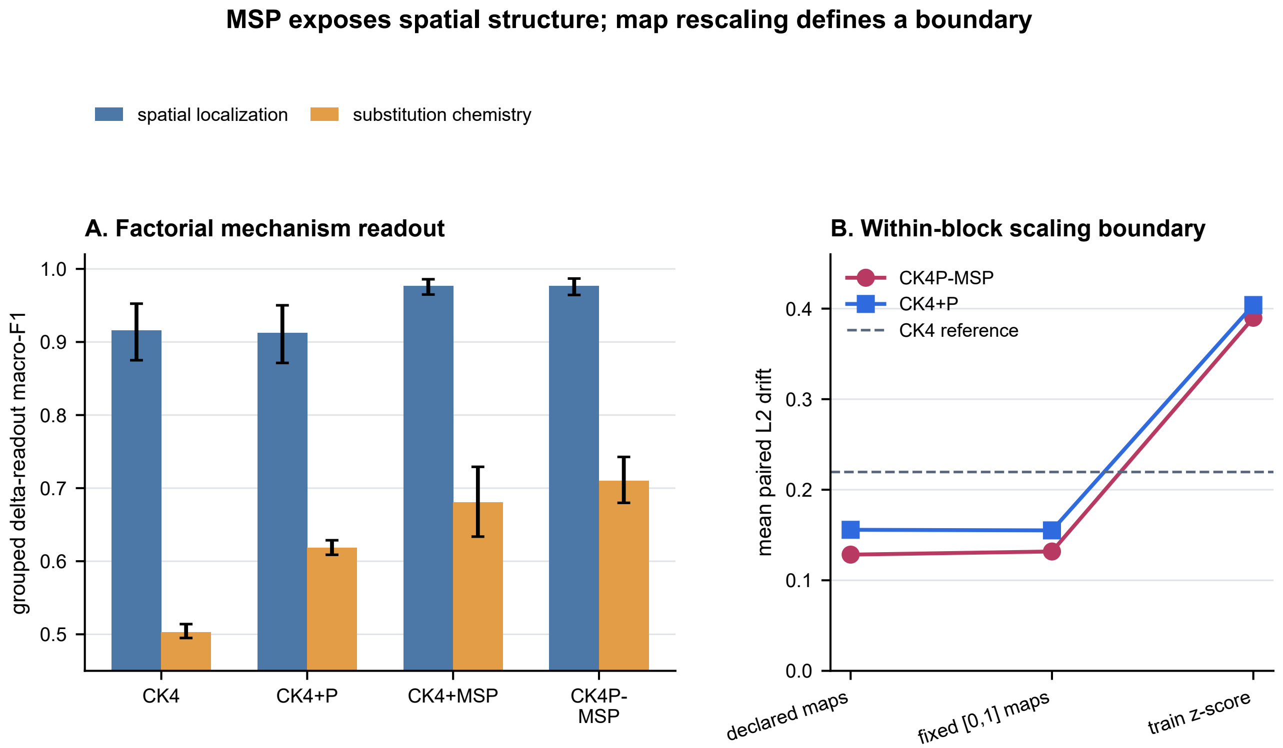

Figure S11 Mechanism-aligned local-change and property-scaling audits. (A) Grouped macro-F1 for spatial localization (contiguous versus dispersed substitutions) and substitution chemistry (property-directed versus random substitutions) across the four K-containing representations. Both tasks used 3,000 template-level observations per representation across 12 length-by-local-mode cells, with all derivatives of a template retained in one fold; error bars are 95% bootstrap intervals across cells. (B) Mean standardized representation drift across 24 WGS length-by-perturbation cells under the declared mappings, fixed per-property $[0, 1]$ mappings and train-fit coordinate z-scores. Higher macro-F1 and lower drift are preferable; z-scoring is a sensitivity analysis rather than an alternative default contract.

Alt text: Two panels showing that MSP improves spatial and chemistry readout, P adds a smaller chemistry increment, and fixed property remapping preserves drift while train-fit z-scoring does not.

Table S4. Prespecified conditional block contrasts in the 2×2 spatial-localization-by-substitution-chemistry audit. Macro-F1 was estimated by template-grouped fivefold cross-validation. Confidence intervals are 10,000-resample paired-cell bootstrap intervals, and q values are Benjamini–Hochberg adjusted across the six contrasts. The matched cells share the controlled template panel; inference is therefore restricted to consistency across this experimental grid.

conditional block	readout target	full	comparator	Δ macro-F1 [95% CI]	BH q
K | P+MSP	spatial localization	0.977	0.964	0.013 [0.001, 0.025]	0.1553
K | P+MSP	substitution chemistry	0.710	0.758	-0.048 [-0.058, -0.037]	0.0009766
P | CK4+MSP	spatial localization	0.977	0.977	0.000 [-0.001, 0.002]	0.5195
P | CK4+MSP	substitution chemistry	0.710	0.680	0.030 [0.013, 0.048]	0.03149
MSP | CK4+P	spatial localization	0.977	0.912	0.065 [0.034, 0.098]	0.0009766
MSP | CK4+P	substitution chemistry	0.710	0.618	0.092 [0.059, 0.130]	0.0009766

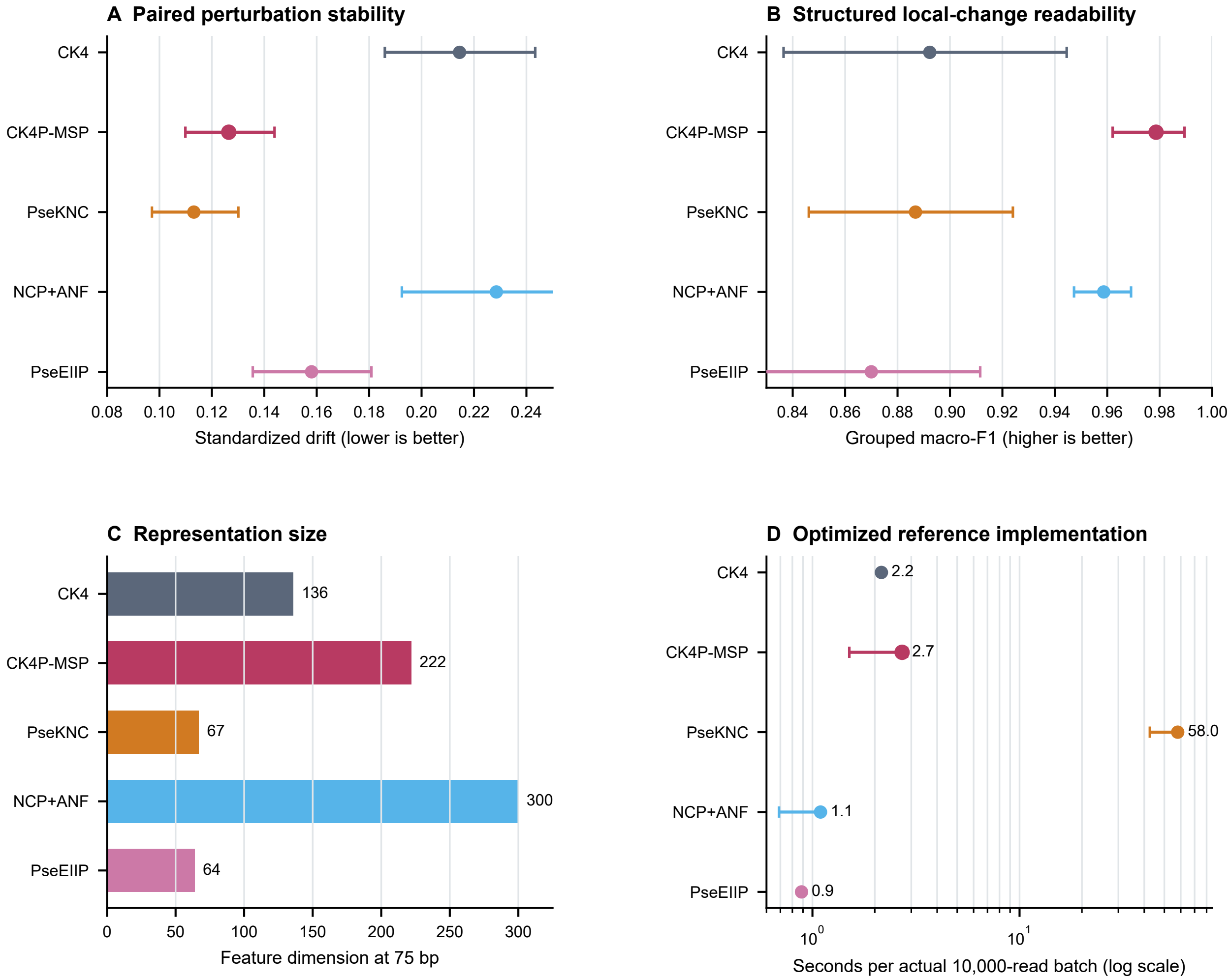

Figure S12 Historical handcrafted descriptor boundary audit. (A) Mean standardized representation drift with 95% matched-cell bootstrap intervals across 30 WGS length-by-perturbation cells; lower is preferable. PseKNC has lower drift than CK4P-MSP. (B) Logistic grouped local-versus-nuisance delta-readout macro-F1 with 95% matched-cell bootstrap intervals across 12 length-by-local-mode cells; all derivatives of a source template remained in one fold, and higher is preferable. CK4P-MSP has the strongest local-change readability in this comparison. These intervals quantify analysis-grid variation, not independent-cohort generalization. (C) Feature dimensions at 75 bp. NCP+ANF scales with the declared read-length budget, whereas the other descriptors are fixed width. (D) Median wall-clock feature-extraction time and interquartile range over five repeats of an actual 10,000-read 75-bp batch, shown on a logarithmic axis. A separate actual 100,000-read pass was used to audit scaling. Methods were warmed up, their order was randomized within each repeat, garbage collection preceded timing and numerical libraries were limited to one thread. Timings were obtained on an Intel Core Ultra 5 125H workstation with 31.6 GB RAM and are implementation-specific. PseKNC used $k = 3$, $\lambda = 3$ and $w = 0.05$; deterministic ambiguous-base extensions are defined in Methods. The figure establishes a stability–readability–dimension boundary rather than a universal ranking.

Alt text: Four panels show that PseKNC has the lowest drift and dimension, CK4P-MSP has the highest grouped local-change readability and near-CK4 optimized runtime, and NCP+ANF is position-explicit but length-dependent in dimension.

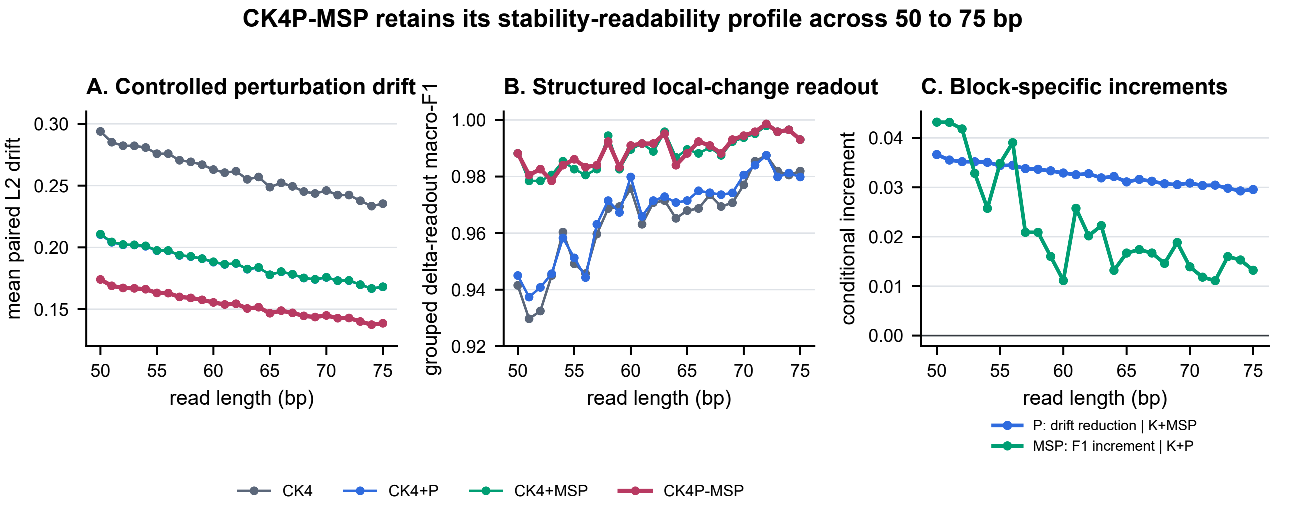

Figure S13 Shared-template continuity audit across every integer length from 50 to 75 bp. (A) Mean paired standardized L2 drift over six controlled perturbations for CK4, CK4+P, CK4+MSP and CK4P-MSP; lower is preferable. Each point summarizes 420 shared source templates that were prefix-truncated from the same 150-bp windows. (B) Mean grouped local-versus-nuisance delta-readout macro-F1 over four local modes with 180 template triplets per length-by-mode cell and fivefold template-grouped cross-validation; higher is preferable. (C) Conditional block-specific increments: P-associated drift reduction is CK4+MSP minus CK4P-MSP, whereas the MSP-associated readability increment is CK4P-MSP minus CK4+P. Positive values indicate improvement on the stated metric. Lines show descriptive analysis-cell means without uncertainty intervals. The dense sweep isolates lower-range length sensitivity; it does not constitute a continuous 50–150 bp scan or a population-level length-response estimate.

Alt text: Three line panels show smooth drift reduction, high grouped local-change readout and positive metric-specific P and MSP increments from 50 to 75 bp.

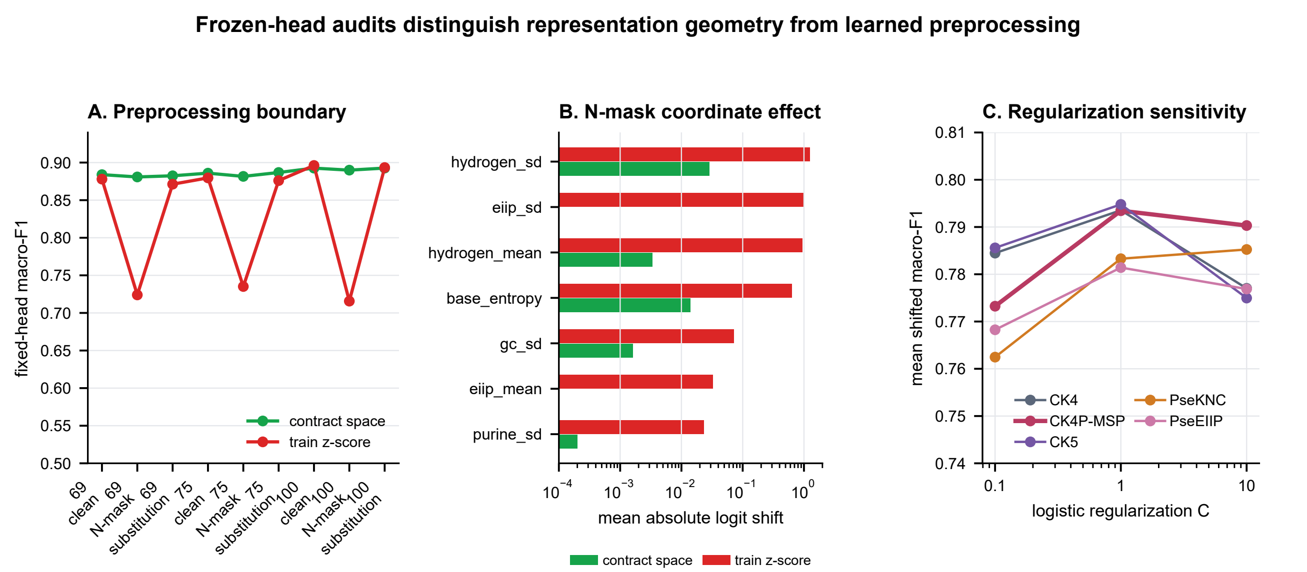

Figure S14 Frozen-head preprocessing and regularization sensitivity. (A) Macro-F1 for CK4P-MSP in the prespecified single CAMI target task. Contract-space probes use the declared row/block-normalized representation directly; train z-score probes apply coordinate-wise means and variances fitted only on the 100-bp clean training fold. Each point aggregates 2,486 held-out source groups through fivefold source-grouped cross-validation. (B) Mean absolute logistic-logit shifts of the seven most affected P coordinates under 100-bp, 3% N masking, averaged over five folds. Green bars denote contract-space contributions and red bars denote amplification after train-fit z-scoring; the horizontal axis is logarithmic. (C) Mean shifted macro-F1 across six CAMI target tasks and eight non-baseline length-by-condition cells per target at $C = 0.1$, 1 and 10. The classifier was otherwise unchanged and was never refitted on shifted conditions. $C = 1$ was prespecified; the other values are sensitivity analyses rather than selected alternatives. These panels show that low representation drift does not guarantee invariance to learned downstream preprocessing.

Alt text: A line panel shows N-mask-specific degradation after train-fit z-scoring, a log-scale bar panel identifies amplified low-variance biochemical coordinates, and a final line panel shows regularization-dependent fixed-head readout.

Table S5. Property-scaling sensitivity across 24 WGS length-by-perturbation cells. The declared mapping is the manuscript contract; fixed $[0, 1]$ maps and train-fit coordinate z-scores test sensitivity to coordinate scaling.

representation	property scaling	paired cosine	L2 drift	retrieval
CK4P-MSP	declared maps	0.989	0.128	1.000
CK4P-MSP	fixed [0,1] maps	0.989	0.132	1.000
CK4P-MSP	train-fit coordinate z-score	0.887	0.390	0.987
CK4+P	declared maps	0.984	0.156	1.000
CK4+P	fixed [0,1] maps	0.984	0.155	1.000
CK4+P	train-fit coordinate z-score	0.869	0.404	0.840

Table S6. Complete P and MSP coordinate definitions. Population SD uses denominator L over the read-level mapped values. MSP uses bin means only in the primary representation.

block	coordinate family	definition	dimensions
P	hydrogen-bond class	Mean and population SD of A/T$=2$, C/G$=3$, N$=0$	2
P	GC indicator	Mean and population SD of C/G$=1$, A/T/N$=0$	2
P	purine indicator	Mean and population SD of A/G$=1$, C/T/N$=0$	2
P	EIIP mapping	Mean and population SD of A$=0.1260$, C$=0.1340$, G$=0.0806$, T$=0.1335$, N$=0$	2
P	ambiguity	Fraction of N bases among A/C/G/T/N characters	1
P	scaled length	Read length divided by 200	1
P	scaled entropy	Shannon entropy over A/C/G/T/N divided by $\log_2 5$	1
MSP	five per-base channels	Hydrogen, GC, purine, EIIP and N-indicator values; mean pooled within each relative-position bin	$5 \times 15 = 75$
MSP	relative-position bins	For each $b \in \{2, 3, 4, 6\}$, edges are $\lfloor jL/b \rfloor$; empty bins use a zero row; the main method excludes within-bin SD	15 bins

Table S7. Public reference assemblies used to sample the controlled WGS read templates. Each species is represented by the listed NCBI assembly accession.

genus	species	NCBI assembly accession
Acinetobacter	*Acinetobacter baumannii*	GCF_900011295.1
Acinetobacter	*Acinetobacter calcoaceticus*	GCF_000368965.1
Acinetobacter	*Acinetobacter nosocomialis*	GCF_041021905.1
Acinetobacter	*Acinetobacter pittii*	GCF_000369045.1
Burkholderia	*Burkholderia cenocepacia*	GCF_000009485.1
Burkholderia	*Burkholderia cepacia*	GCF_001411495.1
Burkholderia	*Burkholderia multivorans*	GCF_000010545.1
Candida	*Candida albicans*	GCF_000182965.3
Candida	*Candida dubliniensis*	GCF_000026945.1
Candida	*Candida parapsilosis*	GCF_000182765.1
Candida	*Candida tropicalis*	GCF_000006335.3
Enterobacter	*Enterobacter asburiae*	GCF_900077725.1
Enterobacter	*Enterobacter bugandensis*	GCF_900324475.1
Enterobacter	*Enterobacter cloacae*	GCF_905331265.2
Enterobacter	*Enterobacter hormaechei*	GCF_000213995.1
Enterobacter	*Enterobacter kobei*	GCF_023023125.1
Escherichia	*Escherichia coli*	GCF_000005845.2
Escherichia	*Escherichia fergusonii*	GCF_000026225.1
Klebsiella	*Klebsiella pneumoniae*	GCF_000240185.1
Klebsiella	*Klebsiella quasipneumoniae*	GCF_003181175.1
Klebsiella	*Klebsiella variicola*	GCF_000025465.1

Table S8. Six-target CAMI_TOY_low fixed-head summary at the prespecified contract-space $C = 1$ probe. Clean F1 is averaged over the six 100-bp clean target tasks. Shifted F1 and retention are first averaged over eight non-baseline length-by-condition cells within each target and then over six targets. Minimum values are minima of the corresponding target-level means, not individual read-level observations. Each task used 1,200 target and 1,200 stratified background source groups with fivefold source-grouped cross-validation. The six tasks share a source pool and are not independent-community replicates.

representation	dimensions	clean F1	shifted F1	minimum target F1	retention	minimum target retention
CK4	136	0.823	0.794	0.693	0.963	0.925
CK4+P	147	0.818	0.794	0.698	0.970	0.936
CK4+MSP	211	0.818	0.793	0.698	0.969	0.933
CK4P-MSP	222	0.816	0.793	0.699	0.971	0.940
CK5	512	0.817	0.795	0.698	0.972	0.953
PseKNC	67	0.803	0.783	0.684	0.974	0.956
PseEIIP	64	0.808	0.781	0.678	0.965	0.931
NCP+ANF	400	0.692	0.661	0.530	0.953	0.897
Hashed $k = 15$	222	0.505	0.505	0.499	1.000	0.979

Table S9. Metadata for the six prespecified CAMI_TOY_low target definitions. Scientific names and ranks were retrieved from NCBI Taxonomy by TaxID. Selection rank reflects abundance among eligible labels after deterministic mate collapse and exclusion of sequences shorter than 100 bp. Each one-versus-background task used 1,200 target and 1,200 stratified background source groups. The tasks share one source pool, and a source read can contribute to the background class of more than one task.

selection rank	NCBI TaxID	scientific name	taxonomic rank	target/background groups
1	434085	*gamma proteobacterium IMCC2047*	species	1,200/1,200
2	247639	*marine gamma proteobacterium HTCC2080*	species	1,200/1,200
3	1050222	*Paenibacillus sp. Aloe-11*	species	1,200/1,200
4	552396	*Erysipelotrichaceae bacterium 5_2_54FAA*	species	1,200/1,200
5	1122939	*Patulibacter americanus DSM 16676*	strain	1,200/1,200
6	1097667	*Patulibacter medicamentivorans*	species	1,200/1,200